

Quiz – 5

Time Allowed: 30

Mins Total Marks: 20

The "DataPlan" Usage Tracker

Instructions: You are tasked with implementing a C++ class that manages mobile data usage for a user.

Part 1: Complete the DataPlan class definition below. Fill in the code exactly where the comments `/* . . . */` instruct you to.

Part 2: Write the `main()` function. Follow the high-level instructions within the `main()` comments to implement the required logic.

Part 3: In the "Predicted Output" section, write the *exact* text that will be printed to the console when your completed `main()` function is executed.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
class DataPlan {
```

```
private:
```

```
    string user;    // The name of the user
```

```
    double dataLimit; // Monthly data limit in GB
```

```
    double dataUsed;  // Data used so far in GB
```

```
    string status;    // Current status: "Active", "Warning", or "Suspended"
```

```
public:
```

```
    // -- Constructor --
```

```
    // 1. Initialize 'user' and 'dataLimit' from parameters.
```

```
    // 2. Set 'dataUsed' to 0.0 and 'status' to "Active".
```

```
// 3. Print "Data plan created for [user] with [dataLimit] GB."
```

```
DataPlan(string userName, double limitGB) {
```

```
    /* YOUR CODE HERE */
```

```
}
```

```
// -- Destructor --
```

```
// Print "Data plan for [user] destroyed."
```

```
~DataPlan() {
```

```
    /* YOUR CODE HERE */
```

```
}
```

```
// -- Getters --
```

```
string getUser() {
```

```
    /* YOUR CODE HERE */
```

```
}
```

```
string getStatus() {
```

```
    /* YOUR CODE HERE */
```

```
}
```

[illegible]

```
void useData(double usageGB) {

    /* YOUR CODE HERE */

}

// -- Reset Data --

// 1. Reset 'dataUsed' to 0.0.

// 2. Set status back to "Active".

// 3. Print "Data usage reset for [user]."
```

```
void resetData() {

    /* YOUR CODE HERE */

}

};

int main() {

    // Create a DataPlan for user "Alice" with a 10.0 GB limit.

    /* YOUR CODE HERE */
```

```
// Use 7.0 GB of data, then 2.0 GB.
```

```
/* YOUR CODE HERE */
```

```
cout << "--- Starting inner scope ---" << endl;
```

```
{
```

```
    // Create a DataPlan for user "Bob" with a 5.0 GB limit.
```

```
    /* YOUR CODE HERE */
```

```
// Use 4.5 GB of data.
```

```
/* YOUR CODE HERE */
```

```
// Print Bob's status.
```

```
/* YOUR CODE HERE */
```

```
} // End of inner scope
```

```
cout << "--- Ending inner scope ---" << endl;
```

```
// Reset Alice's data.
```

```
/* YOUR CODE HERE */
```

```
// Use 1.5 GB of data.
```

```
/* YOUR CODE HERE */
```

```
// Print Alice's final status.
```

```
/* YOUR CODE HERE */
```

```
return 0;
```

```
}Write your output here.
```